

A Brief Note on L^AT_EX

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1 documentclass

```
\documentclass[12pt]{article}
```

Some possible options for the document class:

- article
- report
- book
- letter
- beamer

2 Packages

```
\usepackage[utf8]{inputenc}  
\usepackage{amsmath}  
\usepackage{amsfonts}  
\usepackage{amssymb}  
\usepackage{lipsum}  
\usepackage{xcolor}  
\usepackage{hyperref}
```

3 Meta information

3.1 Title

```
\title{A Brief Note on \LaTeX}
```

3.2 Author

```
\author{Mehmet Hakan Satman}
```

3.3 Date

```
\date{\today}
```

3.4 Making Title

`\maketitle`

4 Table of Contents

`\tableofcontents`

5 Paragraphs

This is the first paragraph. It contains some text to demonstrate how paragraphs are created in `LATEX`.

This is the second paragraph. Notice the space between this paragraph and the previous one, which indicates a new paragraph has started.

This is the third paragraph. It follows immediately after the second paragraph without any additional vertical space.

This is the first paragraph. It contains some text to demonstrate how paragraphs are created in `\LaTeX`.

This is the second paragraph. Notice the space between this paragraph and the previous one, which indicates a new paragraph has started.

`\noindent` This is the third paragraph. It follows immediately after the second paragraph without any additional vertical space.

6 Centering

This text is centered in the document.

`\begin{center}`

This text is centered in the document.

`\end{center}`

7 Quotes

Inline Quote: “This is an inline quote in L^AT_EX.”

‘‘This is an inline quote in \LaTeX.’’

8 Bold, italic, and underline Text

Bold Text: **This text is bold.**

`\textbf{This text is bold.}`

Italic Text: *This text is italic.*

`\textit{This text is italic.}`

Underline Text: This text is underlined.

`\underline{This text is underlined.}`

Bold and Italic Text: ***This text is bold and italic.***

`\textbf{\textit{This text is bold and italic.}}`

Bold and Underline Text: **This text is bold and underlined.**

`\textbf{\underline{This text is bold and underlined.}}`

Bold, italic, and Underline Text: ***This text is bold, italic, and underlined.***

`\textbf{\textit{\underline{This text is bold, italic, and underlined.}}}`

Script Text: THIS TEXT IS IN SMALL CAPITALS.

`\textsc{This text is in small capitals.}`

Computer code: This text is in typewriter font.

`\texttt{This text is in typewriter font.}`

9 Colored Text

9.1 xcolor

Required package: `\usepackage{xcolor}`

Red : **This text is red.**

`\textcolor{red}{This text is red.}`

Blue: **This text is blue.**

`\textcolor{blue}{This text is blue.}`

Green: **This text is green.**

`\textcolor{green}{This text is green.}`

Orange: **This text is orange.**

`\textcolor{orange}{This text is orange.}`

9.2 color

Required package: `\usepackage{color}`

Red : **This text is red.**

`\textcolor{red}{This text is red.}`

Yellow background: **This text has a yellow background.**

`\colorbox{yellow}{This text has a yellow background.}`

Yellow background with red border:

This text has a yellow background with a red border.

`\fcolorbox{red}{yellow}{This text has a yellow background with a red border.}`

10 URLs

Required package: `\usepackage{hyperref}`

`https://www.example.com`

`\url{https://www.example.com}`

11 Sectioning Commands

Section:

`\section{Section Title}`

Subsection:

`\subsection{Subsection Title}`

Subsubsection:

`\subsubsection{Subsubsection Title}`

12 Mathematical Expressions

$$E = mc^2$$

`$$`

`E = mc^2`

`$$`

$$\int_a^b f(x) dx = F(b) - F(a)$$

`$$`

`\int_a^b f(x) \, dx = F(b) - F(a)`

`$$`

$$\Delta = b^2 - 4ac$$

`$$`

`\Delta = b^2 - 4ac`

`$$`

$$\int_0^\infty \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = 1$$

`$$`

`\int_0^\infty \frac{1}{\theta} e^{-\frac{x}{\theta}} \, dx = 1`

`$$`

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

$$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$$

$$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$$

$$\text{sign}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

$$\text{sign}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

$$\text{abs}(x) = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

```


$$\text{abs}(x) = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$


```

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

```


$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$


```

$$\begin{aligned} \max z &= x + 2y \\ \text{Subject to:} \\ 2x + 3y &\leq 52 \\ x + y &\geq 2 \\ x \geq 0, y &\geq 0 \end{aligned} \tag{1}$$

```

\begin{equation}
\begin{split}
\max z &= x + 2y \\
\text{Subject to:} & \\
& 2x + 3y \leq 52 \\
& x + y \geq 2 \\
& x \geq 0, y \geq 0
\end{split}
\end{equation}

```

Order	Symbol	Code
1	α	<code>\alpha</code>
2	β	<code>\beta</code>
3	γ	<code>\gamma</code>
4	δ	<code>\delta</code>
5	Δ	<code>\Delta</code>
6	ϵ	<code>\epsilon</code>
7	ε	<code>\varepsilon</code>
8	γ	<code>\gamma</code>
9	Γ	<code>\Gamma</code>
10	ρ	<code>\rho</code>
11	η	<code>\eta</code>
12	ζ	<code>\zeta</code>
13	θ	<code>\theta</code>
14	λ	<code>\lambda</code>
15	μ	<code>\mu</code>
16	σ	<code>\sigma</code>
17	π	<code>\pi</code>
18	Σ	<code>\Sigma</code>
19	$\int$	<code>\int</code>
20	$\sum$	<code>\sum</code>
21	$\frac{a}{b}$	<code>\frac{a}{b}</code>
22	$\sqrt{x}$	<code>\sqrt{x}</code>
23	$\sqrt[n]{x}$	<code>\sqrt[n]{x}</code>
24	∞	<code>\infty</code>
25	x_i	<code>x_i</code>
26	x^n	<code>x^n</code>
27	x_i^n	<code>x_i^n</code>

Table 1: Common Mathematical Symbols in L^AT_EX

13 Lorem Ipsum

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique

Order	Symbol	Code
1	$\pm$	<code>\pm</code>
2	$\mp$	<code>\mp</code>
3	$\times$	<code>\times</code>
4	$\div$	<code>\div</code>
5	$\cdot$	<code>\cdot</code>
6	$\leq$	<code>\leq</code>
7	$\geq$	<code>\geq</code>
8	$\neq$	<code>\neq</code>
9	$\approx$	<code>\approx</code>
10	$\forall$	<code>\forall</code>
11	$\exists$	<code>\exists</code>
12	$\in$	<code>\in</code>
13	$\notin$	<code>\notin</code>
14	$\subset$	<code>\subset</code>
15	$\subseteq$	<code>\subseteq</code>
16	$\cup$	<code>\cup</code>
17	$\cap$	<code>\cap</code>
18	$\setminus$	<code>\setminus</code>
19	$\emptyset$	<code>\emptyset</code>
20	$\mathcal{R}$	<code>\mathcal{R}</code>
21	$\mathbb{R}$	<code>\mathbb{R}</code>
22	$\mathbb{N}$	<code>\mathbb{N}</code>
23	$\mathbb{Z}$	<code>\mathbb{Z}</code>
24	$\mathbb{Q}$	<code>\mathbb{Q}</code>
25	$\mathbb{C}$	<code>\mathbb{C}</code>
26	$\dots$	<code>\ldots</code>

Table 2: Common Mathematical Symbols in L^AT_EX(Continued)

senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget

orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

`\lipsum[1-2]`

14 Tables

14.1 Tabular Environment

A	B	C
1	2	3
4	5	6

```
\begin{tabular}{|c|c|c|}  
\hline  
A & B & C \\  
\hline  
1 & 2 & 3 \\  
4 & 5 & 6 \\  
\hline  
\end{tabular}
```

14.2 Tables with Caption

X	Y	Z
7	8	9
10	11	12

Table 3: A Simple Table

```

\begin{table}[h!]
\centering
\begin{tabular}{|c|c|c|}
\hline
X & Y & Z \\
\hline
7 & 8 & 9 \\
10 & 11 & 12 \\
\hline
\end{tabular}
\caption{A Simple Table}
\end{table}

```

15 Listings

15.1 Itemize Environment

- First item
- Second item
- Third item

```

\begin{itemize}
\item First item
\item Second item
\item Third item
\end{itemize}

```

15.2 Enumerate Environment

1. First item
2. Second item
3. Third item

```

\begin{enumerate}
\item First item

```

```

\item Second item
\item Third item
\end{enumerate}

```

15.3 Text in boxes

This is a text box in L^AT_EX. You can use it to highlight important information or to create a visual separation from the surrounding text.

```

\begin{center}
\framebox{
  \begin{minipage}{0.8\textwidth}
    This is a text box in \LaTeX. You can use it to highlight
    important information or to create a visual
    separation from the surrounding text.
  \end{minipage}
}
\end{center}

```

16 Referencing

16.1 Equation Referencing

$$\Delta = b^2 - 4ac \tag{2}$$

```

\begin{equation}
\Delta = b^2 - 4ac
\label{eq:delta}
\end{equation}

```

Referencing: Equation 2 shows the discriminant of a quadratic equation.

Referencing: Equation `\ref{eq:delta}` shows the discriminant of a quadratic equation.

A	B	C
1	2	3
4	5	6

Table 4: A Simple Table

16.2 Table Referencing

```
\begin{table}[h!]
\centering
\begin{tabular}{|c|c|c|}
\hline
A & B & C \\
\hline
1 & 2 & 3 \\
4 & 5 & 6 \\
\hline
\end{tabular}
\caption{A Simple Table}
\label{tab:simple}
\end{table}
```

Table 4 shows a simple table with three columns and two rows of data.

Table `\ref{tab:simple}` shows a simple table with three columns and two rows of data.

17 Presentations

Required package: `beamer`

Example:

```
\documentclass{beamer}
\begin{document}
\begin{frame}
\frametitle{Sample Slide}
This is a sample slide in a Beamer presentation.
\end{frame}
\end{document}
```

Changing the theme:

```
\documentclass{beamer}
\usetheme{Madrid}
\begin{document}
\begin{frame}
\frametitle{Sample Slide with Theme}
This is a sample slide in a Beamer presentation with the Madrid theme.
\end{frame}
\end{document}
```

Some possible themes for Beamer presentations include:

- Madrid
- Berlin
- Copenhagen
- CambridgeUS
- Frankfurt